

Statistics 6910: Graphics 1

Why are graphics discussed in this course?

- Good graphics today require the computer
- Visualization enters every step of the data analysis cycle
 - Data cleaning – are there anomalies?
 - Exploration
 - Model checking
 - Reporting results
- Plots can uncover structure in data that can't be detected with numerical summaries
- Important communication skill

Preview: Statistics 6560

- Advanced graphics discussed in Stat 6560 (Graphical Methods)
- Topics include:
 - Bad graphics (and how to avoid those)
 - History of graphics
 - Human perception & color
 - From 1-dim (categorical) data to high-dim data
 - Statistical maps
 - Interactive and dynamic graphics

R's graphics model

- There are two models in R – painter and object-oriented
- We will use the painter's model
- The other is easy to get started but hard to tweak
- Painter's model – start with a blank canvas, add/paint on it in multiple passes

Know your data types

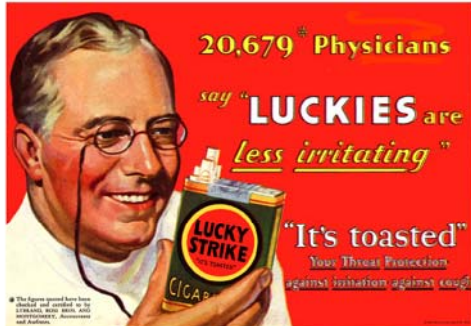
The appropriate graphical techniques depend on the kind of data that you are working with

- **Quantitative**
 - **continuous** – e.g. height, weight
 - **discrete** – numeric data with few values, e.g., number of children in family
- **Qualitative (= Categorical)**
 - **ordered** – categories with an order but no meaningful distance between, e.g., number of stars for a movie rating
 - **nominal** – categories have no meaningful order, e.g., race

Case: Infant Health

```
load(url("http://www.math.usu.edu/~symanzik/teaching/2013_stat6910/LectureNotes/KaiserBabies.rda"))
```

Smoking:
Some doctors used to recommend it



Today:

SURGEON GENERAL'S WARNING: Smoking Causes Lung Cancer, Heart Disease, Emphysema, and May Complicate Pregnancy.

Kaiser Study

- Oakland Kaiser mothers
- 1960s
- Measure the babies weight (in ounces) at birth
- All babies:
 - Male
 - Single births (no twins, etc.)
 - Survived 28 days

Information collected on mother's and their babies

- Birth weight (ounces)
- Gestation (weeks)
- Parity - total number of previous pregnancies
- Mother's height and weight
- Mother's smoking status
- Mother's age, race, education level, income
- And more...

Distribution of Birth Weight

- The **distribution** is the pattern of variation in the birth weights.
- It provides the numerical values for birth weight and how often each value occurs.
- A **histogram/density plot** shows the shape of the distribution

Histograms

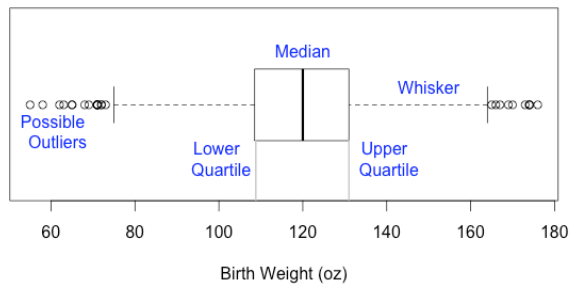
- Are a special case of density plots
- AREA = Proportion (or percent)
- The area of a bar:

$$\text{Height} * \text{Width} = \text{Area}$$

$$(\text{Proportion/oz}) * \text{oz} = \text{Proportion}$$
- Histograms are not the same as bar charts
- With bar charts, it is only the height that matters. **Bar charts are for qualitative data!**

Boxplot:

```
boxplot(infants$bwt)
```

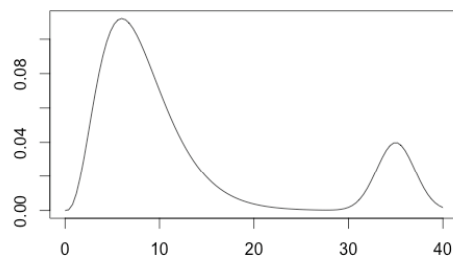


```
boxplot(infants$bwt, ...)
```

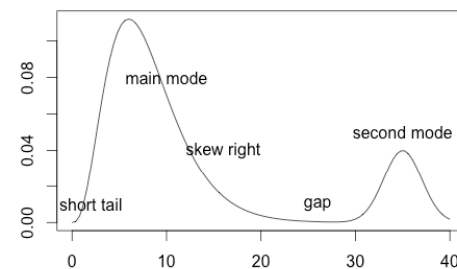
Looking for Structure: Quantitative Distribution

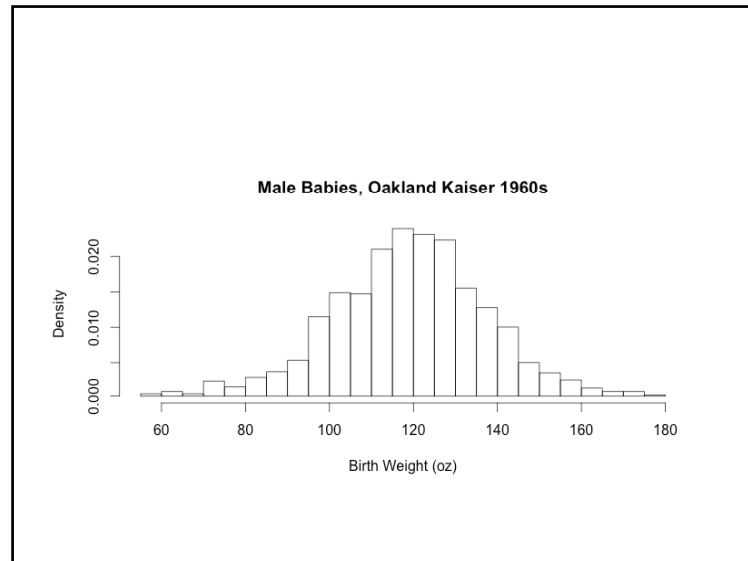
- **Distribution:** pattern of values for a variable
- **Mode:** high density region
- **Long Tail:** many observations far from center
- **Symmetry/Skewness:** distribution of values the left and right of the center.
- **Gaps:** places where there are no observations.
- **Outliers:** unusually large or small values that falls well beyond the overall pattern of data

What Structure Do You See?



What Structure Do You See?





Case: College Students

```
load(url("http://www.math.usu.edu/
~symanzik/teaching/2013_stat6910/
LectureNotes/videogame.rda"))
```

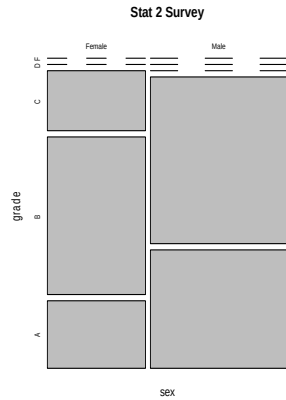
STAT 2 Survey

- Random Sample of 91 of 314 Cal students enrolled in Stat 2
- Survey collected the following info:
 - sex – Male/Female
 - grade – grade expected in the course ("A", "B", "C", "D", "F")
- What type of data are these?
 - sex is qualitative (nominal)
 - grade is qualitative with an ordering

Method of Comparison

- Often, we not only want to better understand a distribution, but we want to compare the distribution for subgroups or to compare against another population or standard
- How do you think the expected grade distribution might vary with gender?

Two Qualitative Variables



```
mosaicplot(table(video$sex, video$grade),
main="Stat 2 Survey")
```

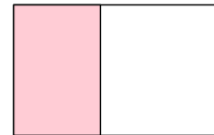
How to read a Mosaic Plot

There are 91
students in the
survey.
Think of them as
spread out evenly
in the box



New Plot: Mosaic Plot

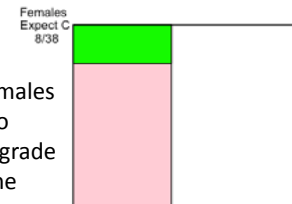
Put all the
females on
one side of
the box.
There are 38.



Females 38/91

New Plot: Mosaic Plot

Rearrange the females
so that those who
expect the same grade
are together in the
box.
8 of the 38 expect a C



Females 38/91

